

Crib Sentry

The baby monitor that alerts instead of only recording.

Tier 2

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The problem

A traditional baby monitor only helps if someone is watching it. Parents can't watch continuously, so a lot of what the camera records goes unnoticed.

The goal is to cut down on what gets missed as much as possible.

What the system reports

All clear

Left the crib

Not visible

Hazard present

Who cares

Parents and caregivers of infants and toddlers.

Why it matters

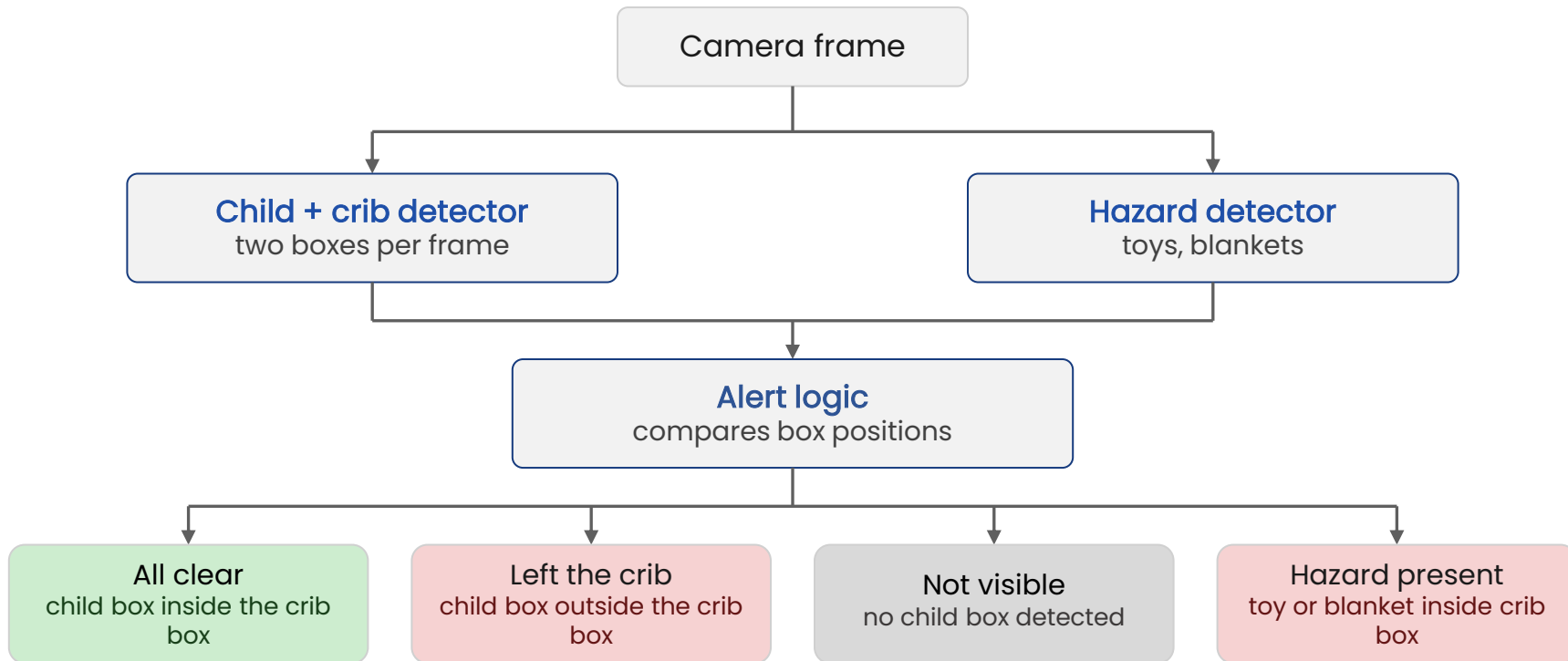
A feed nobody is monitoring provides no protection.

The solution

A system that monitors the feed and alerts only when a defined condition occurs.

The solution

One frame from a crib camera goes in; two detectors run; application logic compares the boxes and returns one of four states.



Technical approach

CV TECHNIQUE

Object detection

Draws boxes around the child, crib, and hazards so the logic can compare them.

ARCHITECTURE

CNN

YOLO11 is a convolutional neural network, the standard model type for images.

HOW IT IS USED

Transfer learning

Starts from a model already trained on COCO, then trained more on my images.

FRAMEWORK

PyTorch through Ultralytics

Free and open source.

Why this fits. Every alert depends on where one box is compared to another, so bounding boxes are what I need. YOLO11 can be trained on a few hundred images on a free Colab GPU, and it can run on video later without changing the approach. During training, I'll also add darker, blurred, and rotated copies of the images so the model does better on real nursery footage.

Data plan

Source	Size	Labels	License
Baby in Crib (Roboflow)	242 images 169 / 49 / 24	Child, Crib, already labeled	CC BY 4.0
CribHD T + B (Northeastern)	1,369 training images	hard-toy, soft-toy, blanket, converted from outlines to boxes	Non-commercial
CribHD-C	120 images	Unlabeled, held out for testing	Non-commercial

Why CribHD-C is being held out

The Baby in Crib images are studio stock photos. CribHD-C looks more like real overhead camera footage, so I'm holding it out to see how much worse the system does on it.

Known gap

No dataset I'm using has a child outside a crib, which is half of the alert logic.

Success metrics

These are my targets. At the Final, I'll show my actual results next to them.

PRIMARY

85%

how often the alert is correct on the 24
test images

SECONDARY

< 1 sec

per image, start to finish

Why alert accuracy, not mAP. mAP only measures how well the detector finds boxes. Alert accuracy measures whether the final alert is right, which is what a parent actually sees. I care more about recall than precision. For a safety system, missing a real problem is worse than a false alarm.

Milestone plan

Weeks counted from the start of term. Blueprint is week 6, Final is week 9.

Phase	Goal	Milestone	Week
Blueprint	Plan approved	Midterm submitted	6
First Working Demo	Pretrained model runs start to finish on a few sample images	Something works, even if rough	7
Make It Yours	Add data, training, and alert logic	System works on my problem	7–8
Improve and Measure	Test, fix, and measure against the metrics	Metrics recorded	8
Package and Present	Demo video, README, final slides	Final submitted	9

Target: Have the system working by week 8, so week 9 is only for the demo video, README, and slides.

Risks and resources

Occlusion **Probability: High**

A child lying down or partly under a blanket might not be detected, since the training images only show children sitting or standing and fully visible.

Plan B: If the crib is detected but no child is found in it, report "not visible" instead of "left the crib." It's better to say it can't tell than to give a wrong alert.

Missing condition **Probability: High**

No dataset has a child outside a crib, which is half of the alert logic.

Plan B: Find a small supplementary dataset of children outside cribs on Roboflow Universe.

Compute

Google Colab free tier with a T4 GPU. My PC's GTX 1660 as a backup.

Cost: \$0

Every model, dataset, and tool I'm using is free or open source.